Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Advanced Java (01CT1623)	**Aim:** Take Home Exam 1 -JDBC	
Take Home Exam 1	**Date :-21-04-2026**	**Enrollment No:-** 92301733024

Model/Book.java

```
package model;

public class Book {
    private int bookId;
    private String title;
    private String author;
    private int quantity;

    public Book() {}

    public Book(String title, String author, int quantity) {
        this.title = title;
        this.author = author;
        this.quantity = quantity;
    }

    public Book(int bookId, String title, String author, int quantity) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.quantity = quantity;
    }

    // Getters & Setters
    public int getBookId() { return bookId; }
    public void setBookId(int bookId) { this.bookId = bookId; }

    public String getTitle() { return title; }
    public void setTitle(String title) { this.title = title; }

    public String getAuthor() { return author; }
    public void setAuthor(String author) { this.author = author; }

    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity = quantity; }
}
```

util/DBConnection:

```
package util;

import java.sql.Connection;
import java.sql.DriverManager;
```

```
public class DBConnection {

    private static final String URL = "jdbc:mysql://localhost:3306/librarydb";
    private static final String USER = "root";
    private static final String PASSWORD = "";

    public static Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

dao/BookDAO.java

```
package dao;

import model.Book;
import util.DBConnection;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class BookDAO {

    // ADD BOOK
    public void addBook(Book book) {
        String sql = "INSERT INTO books(title, author, quantity) VALUES (?, ?, ?)";

        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, book.getTitle());
            ps.setString(2, book.getAuthor());
            ps.setInt(3, book.getQuantity());

            ps.executeUpdate();
            System.out.println("Book added successfully!");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // VIEW ALL BOOKS
```

```
public List<Book> getAllBooks() {
    List<Book> list = new ArrayList<>();
    String sql = "SELECT * FROM books";

    try (Connection con = DBConnection.getConnection();
         PreparedStatement ps = con.prepareStatement(sql);
         ResultSet rs = ps.executeQuery()) {

        while (rs.next()) {
            list.add(new Book(
                    rs.getInt("bookId"),
                    rs.getString("title"),
                    rs.getString("author"),
                    rs.getInt("quantity")
            ));
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

// SEARCH BOOK
public List<Book> searchBook(String keyword) {
    List<Book> list = new ArrayList<>();
    String sql = "SELECT * FROM books WHERE title LIKE ? OR author LIKE ?";

    try (Connection con = DBConnection.getConnection();
         PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setString(1, "%" + keyword + "%");
        ps.setString(2, "%" + keyword + "%");

        ResultSet rs = ps.executeQuery();

        while (rs.next()) {
            list.add(new Book(
                    rs.getInt("bookId"),
                    rs.getString("title"),
                    rs.getString("author"),
                    rs.getInt("quantity")
            ));
        }

    } catch (Exception e) {
```

```
            e.printStackTrace();
        }
        return list;
    }

    // UPDATE BOOK
    public void updateBook(int id, int quantity) {
        String sql = "UPDATE books SET quantity=? WHERE bookId=?";

        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, quantity);
            ps.setInt(2, id);

            ps.executeUpdate();
            System.out.println("Book updated!");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // DELETE BOOK
    public void deleteBook(int id) {
        String sql = "DELETE FROM books WHERE bookId=?";

        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, id);
            ps.executeUpdate();
            System.out.println("Book deleted!");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

main/MainApp.java

```
package main;

import dao.BookDAO;
import model.Book;
```

```
public class MainApp {
    public static void main(String[] args) {

        BookDAO dao = new BookDAO();

        // ADD
        dao.addBook(new Book("Java Programming", "James Gosling", 10));

        // VIEW
        dao.getAllBooks().forEach(b ->
                System.out.println(b.getBookId() + " " + b.getTitle())
        );

        // SEARCH
        dao.searchBook("Java").forEach(b ->
                System.out.println("Found: " + b.getTitle())
        );

        // UPDATE
        dao.updateBook(1, 20);

        // DELETE
        dao.deleteBook(2);
    }
}
```

main/MainApp.java:

```
package main;

import dao.BookDAO;
import model.Book;

public class MainApp {
    public static void main(String[] args) {

        BookDAO dao = new BookDAO();

        // ADD
        dao.addBook(new Book("Java Programming", "James Gosling", 10));

        // VIEW
        dao.getAllBooks().forEach(b ->
                System.out.println(b.getBookId() + " " + b.getTitle())
```

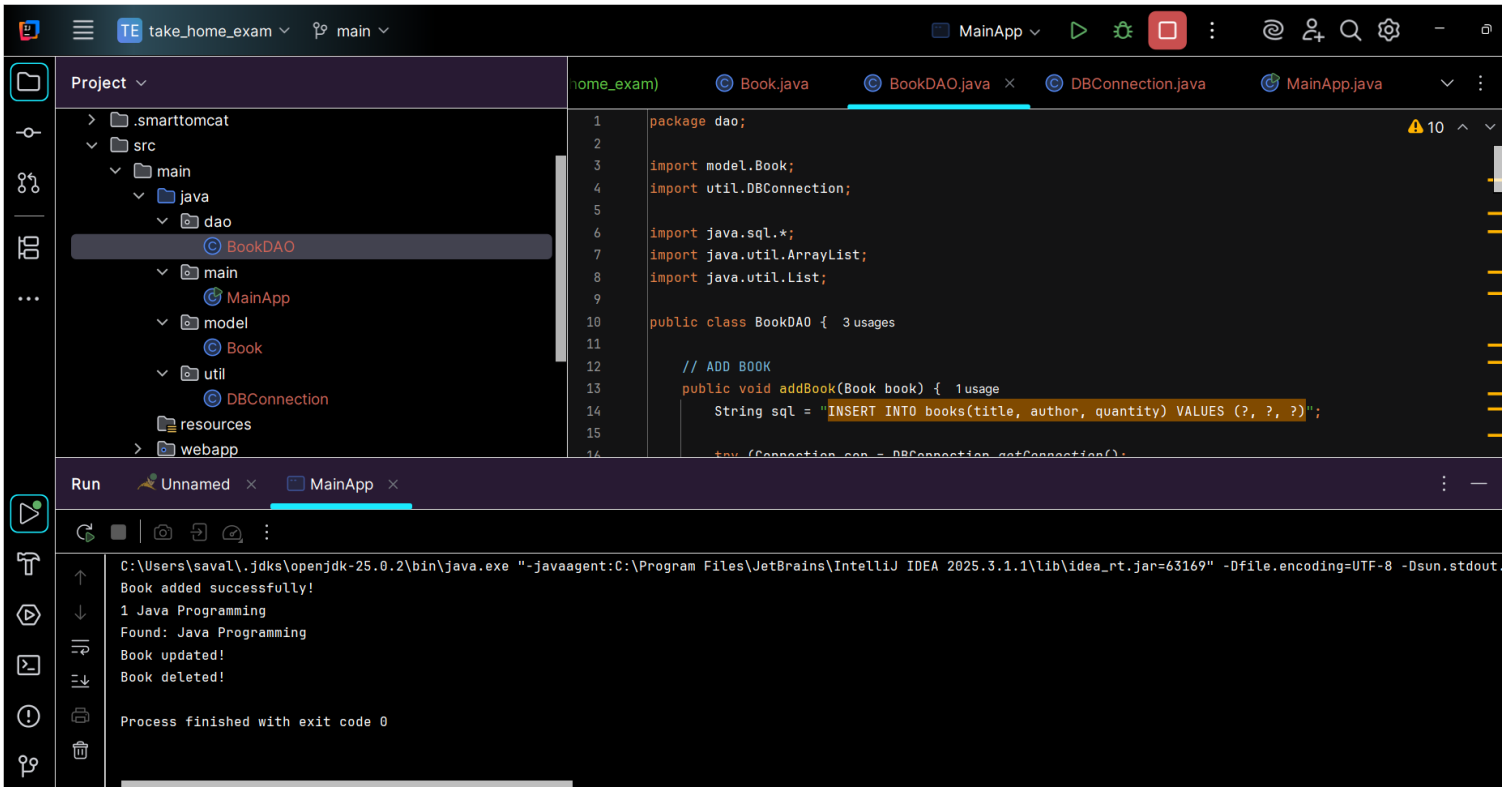
```
        );

        // SEARCH
        dao.searchBook("Java").forEach(b ->
            System.out.println("Found: " + b.getTitle())
        );

        // UPDATE
        dao.updateBook(1, 20);

        // DELETE
        dao.deleteBook(2);
    }
}
```

OUTPUT:



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←T→	bookId	title	author	quantity
☐ Edit Copy Delete	1	Java Programming	James Gosling	20

☐ Check all *With selected:* Edit Copy Delete Export